

UniChat Enterprise SaaS Platform

Next-Generation Multi-Tenant Academic Communication Ecosystem

Software Requirements Specification (SRS)

IEEE 830 Inspired Enterprise Edition

Document Control

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System Type	Multi-Tenant SaaS Academic Communication Platform

1. Executive Overview

1.1 Project Vision

UniChat Enterprise is a cloud-native, multi-tenant academic communication ecosystem engineered to modernize institutional communication infrastructure across East Africa.

The platform combines:

- Real-time messaging
- Academic collaboration
- Institution governance
- AI-powered assistance

- White-label SaaS deployment
- Enterprise-grade scalability

into a single unified platform.

Unlike traditional messaging applications designed primarily for consumer use, UniChat Enterprise is architected specifically for academic institutions requiring:

- role-aware communication,
- centralized governance,
- institutional branding,
- auditability,
- structured academic hierarchy,
- and secure multi-tenant isolation.

The system adopts modern distributed-system principles inspired by large-scale platforms such as WhatsApp, Discord, Slack, Microsoft Teams, and Telegram.

2. Strategic Product Positioning

2.1 Product Identity

UniChat Enterprise is not positioned as:

“A WhatsApp clone for schools.”

The platform is positioned as:

“An Academic Communication Operating System.”

The system acts as the institutional digital communication backbone supporting:

- universities,
 - colleges,
 - technical institutes,
 - training centers,
 - and large educational organizations.
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3. Core Product Objectives

3.1 Primary Objectives

The platform aims to:

1. Replace fragmented academic communication channels.
 2. Eliminate dependence on unmanaged consumer messaging apps.
 3. Provide institution-controlled messaging infrastructure.
 4. Support scalable academic collaboration.
 5. Deliver secure and isolated multi-tenant SaaS deployment.
 6. Enable white-label institutional branding.
 7. Operate efficiently within East African network conditions.
 8. Support AI-assisted educational communication.
 9. Enable enterprise-grade observability and governance.
 10. Create a commercially scalable EdTech SaaS platform.
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4. System Architecture Philosophy

4.1 Architectural Principles

UniChat Enterprise follows these architectural principles:

Principle	Description
Event-Driven Design	Internal services communicate asynchronously through message queues
Multi-Tenant Isolation	Each institution operates within logically isolated boundaries
Horizontal Scalability	Infrastructure scales by adding nodes rather than upgrading servers
Fault Tolerance	Failure of one subsystem should not collapse the platform
Real-Time First	Messaging and collaboration optimized for low latency
API-Centric Design	All services expose structured APIs
Cloud-Native Architecture	Designed for containerized deployment
Security by Design	Authentication, encryption, and auditing embedded from inception
Offline Resilience	Functional under unstable African network environments

5. High-Level Enterprise Architecture

5.1 Distributed Architecture Overview

The platform is built using a distributed microservice architecture.

Core Layers

Layer	Responsibility
Client Layer	Flutter mobile apps and React web applications
API Gateway Layer	Centralized routing, rate limiting, security
Service Layer	Independent business-domain microservices
Event Streaming Layer	Asynchronous communication and decoupling
Real-Time Layer	WebSocket communication and presence
Data Layer	Persistent storage, caching, and indexing
Infrastructure Layer	Containers, orchestration, monitoring

6. Recommended Technology Stack

6.1 Enterprise Technology Stack

Category	Technology
Mobile Application	Flutter
Web Frontend	React.js + Next.js
Backend Runtime	Node.js 20 LTS
API Framework	Fastify
API Gateway	NGINX + Kong Gateway
Real-Time Communication	Socket.IO
Voice/Video Calls	WebRTC + mediasoup
Event Streaming	Apache Kafka
Background Jobs	RabbitMQ
Database	PostgreSQL 15

Category	Technology
Search Engine	Elasticsearch
Cache Layer	Redis
Object Storage	Cloudflare R2
CDN	Cloudflare
Authentication	JWT + Refresh Tokens
Push Notifications	Firebase Cloud Messaging
SMS Provider	Africa's Talking
Email Provider	SendGrid
Monitoring	Prometheus + Grafana
Centralized Logging	Loki + Grafana
Containerization	Docker
Orchestration	Kubernetes
CI/CD	GitHub Actions
Infrastructure Hosting	Hetzner / DigitalOcean

7. Microservice Architecture

7.1 Service Decomposition

The platform is decomposed into independent domain-oriented services.

Service	Primary Responsibility
Authentication Service	OTP, JWT issuance, sessions
User Service	User profiles and identity
Tenant Service	Institution isolation and branding
Messaging Service	Chat delivery and persistence
Presence Service	Online status and typing indicators
Notification Service	Push, SMS, and email delivery
Media Service	Uploads, compression, transcoding

Service	Primary Responsibility
Analytics Service	Usage metrics and dashboards
Search Service	Full-text and semantic search
AI Service	Moderation, summaries, transcription
Call Service	Voice/video conferencing
Billing Service	Subscription management
Audit Service	Immutable audit logging
Community Service	Academic structure and channels

8. Event-Driven Messaging Architecture

8.1 Messaging Flow

The platform uses asynchronous event streaming.

Example Workflow

1. User sends message.
2. Flutter client emits WebSocket event.
3. API Gateway validates JWT.
4. Messaging Service receives payload.
5. Message event published to Kafka.
6. Independent consumers process events.

Event Consumers

Consumer	Responsibility
Notification Consumer	Push notifications
Analytics Consumer	Usage tracking
Moderation Consumer	AI moderation checks
Audit Consumer	Immutable logging
Persistence Consumer	Database storage
Search Consumer	Elasticsearch indexing

This architecture minimizes coupling and improves scalability.

9. Multi-Tenant SaaS Architecture

9.1 Tenant Isolation

UniChat Enterprise implements secure logical tenant isolation.

Isolation Strategy

- Shared infrastructure
- Tenant-aware routing
- Tenant-specific branding
- Row-Level Security (RLS)
- Tenant-scoped JWT claims
- Per-tenant storage quotas
- Per-tenant feature flags

Each institution operates independently while sharing platform infrastructure.

10. Authentication and Identity Management

10.1 Authentication Flow

Authentication uses institution-scoped OTP verification.

Flow

1. User enters Student ID or Staff ID.
 2. Tenant context is identified.
 3. OTP generated securely.
 4. OTP delivered via SMS or email.
 5. OTP verified.
 6. Access token issued.
 7. Refresh token issued.
 8. Session registered.
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10.2 Security Enhancements

Advanced Security Features

- Refresh token rotation
- Device fingerprinting
- Session revocation

- Multi-device management
 - Suspicious login detection
 - IP anomaly monitoring
 - Brute-force protection
 - Geo-based session analysis
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11. Real-Time Communication Layer

11.1 Messaging Features

Private Messaging

- Text messaging
 - Message reactions
 - Read receipts
 - Reply threading
 - Forwarding
 - Voice notes
 - Media sharing
 - Message editing
 - Delete for everyone
 - Bookmarking
 - Scheduled messages
 - AI-assisted writing
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11.2 Group Messaging

Group Features

- Up to 1,024 members
 - Admin moderation
 - Announcement mode
 - Join approvals
 - Role-based permissions
 - Polls
 - Shared media gallery
 - Academic resource sharing
 - Attendance tracking
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11.3 Presence System

Presence Features

- Online status
 - Typing indicators
 - Last seen
 - Active session tracking
 - Multi-device synchronization
 - Redis-based presence synchronization
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12. Academic Structure Engine

12.1 Institutional Hierarchy

The platform mirrors real academic structures.

Level	Example
Institution	ATC
Faculty	School of Engineering
Department	Computer Science
Cohort	Year 2 CS
Course Group	CS301 Database Systems
Club	Coding Society
Channel	Exam Announcements

12.2 Smart Enrollment Engine

Students are automatically assigned to:

- departments,
- cohorts,
- courses,
- academic groups,
- and announcement channels

using institutional enrollment datasets.

13. Voice and Video Infrastructure

13.1 Calling Architecture

The platform uses:

- WebRTC,
- STUN/TURN servers,
- and mediasoup SFU architecture.

Features

Feature	Supported
Voice Calls	Yes
Video Calls	Yes
Screen Sharing	Yes
Presenter Mode	Yes
Noise Suppression	Yes
Adaptive Bitrate	Yes
Group Calls	Up to 32
Call Recording	Premium Plans

14. AI-Powered Capabilities

14.1 AI Integration Layer

UniChat Enterprise integrates AI-enhanced academic workflows.

AI Features

Feature	Purpose
AI Writing Assistant	Academic message drafting
Lecture Summarization	Summarize lecture discussions
Voice Transcription	Convert voice notes to text
AI Moderation	Detect abuse or harmful content

Feature	Purpose
Semantic Search	Intelligent search results
Meeting Notes	Automatic summaries
Smart Notifications	Priority-aware alerts
Translation Assistance	English/Swahili support

15. Media Infrastructure

15.1 Object Storage Architecture

The system uses an abstraction-based media layer supporting:

- Cloudflare R2
- AWS S3
- MinIO
- Cloudinary compatibility

Media Features

- Automatic compression
- Signed URLs
- Media transcoding
- Thumbnail generation
- Adaptive upload quality
- CDN delivery
- Storage lifecycle policies

16. Search Infrastructure

16.1 Enterprise Search Engine

The platform integrates Elasticsearch for:

- message search,
- semantic retrieval,
- user search,
- file indexing,
- and analytics queries.

Search Features

- Full-text search
 - Fuzzy matching
 - Semantic ranking
 - Date filtering
 - File search
 - Multi-language indexing
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17. Notification Infrastructure

17.1 Notification Channels

Channel	Purpose
Push Notifications	Real-time alerts
In-App Notifications	Activity center
SMS	Critical alerts
Email	Digests and reports
Broadcast Notifications	Institution-wide communication

18. Administrative Infrastructure

18.1 Institution Admin Features

Institution administrators can:

- manage users,
 - manage departments,
 - configure branding,
 - control permissions,
 - analyze platform metrics,
 - schedule broadcasts,
 - manage subscriptions,
 - and configure academic structures.
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18.2 Super Admin Features

Super Admins can:

- manage tenants,
 - control platform-wide settings,
 - configure infrastructure,
 - monitor analytics,
 - manage billing,
 - deploy white-label builds,
 - and oversee global governance.
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19. Observability and Monitoring

19.1 Monitoring Stack

The platform integrates:

- Prometheus,
- Grafana,
- Loki,
- and distributed tracing.

Metrics Monitored

Metric	Purpose
WebSocket latency	Real-time performance
Message throughput	Traffic analysis
API latency	Backend performance
CPU utilization	Infrastructure monitoring
Redis memory usage	Cache monitoring
Kafka lag	Event-stream monitoring
Error rates	Stability analysis

20. Security Architecture

20.1 Enterprise Security Controls

Security Measures

- TLS 1.3 encryption
 - JWT RS256 signing
 - End-to-End Encryption
 - PostgreSQL Row-Level Security
 - Rate limiting
 - IP throttling
 - Signed media URLs
 - Audit logging
 - Immutable event tracking
 - API gateway validation
 - Security headers
 - WAF integration
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21. Scalability Strategy

21.1 Horizontal Scalability

The platform scales horizontally using:

- Kubernetes orchestration,
 - stateless services,
 - Redis clustering,
 - Kafka partitioning,
 - PostgreSQL replication,
 - CDN acceleration,
 - and distributed WebSocket infrastructure.
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22. Offline-First Mobile Strategy

22.1 Offline Resilience

The mobile application supports:

- offline message composition,
- queued delivery,

- local caching,
- retry synchronization,
- and degraded-network optimization.

This is optimized specifically for East African mobile network conditions.

23. White-Label SaaS Model

23.1 White-Label Branding

Each institution may customize:

- app name,
 - colors,
 - logos,
 - splash screens,
 - domains,
 - notification branding,
 - and mobile app identity.
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24. Subscription Model

24.1 SaaS Plans

Plan	Target Market
Starter	Small colleges
Standard	Medium institutions
Premium	Large universities
Enterprise	Multi-campus organizations

25. Deployment Architecture

25.1 Infrastructure Deployment

Deployment Components

Component	Deployment Method
Services	Docker containers
Orchestration	Kubernetes
Reverse Proxy	NGINX
CDN	Cloudflare
Database	PostgreSQL clusters
Monitoring	Prometheus/Grafana
Logging	Loki

26. API Architecture

26.1 REST + WebSocket APIs

REST Endpoints

Endpoint	Purpose
/api/v1/auth	Authentication
/api/v1/users	User management
/api/v1/messages	Messaging
/api/v1/groups	Group management
/api/v1/channels	Broadcast channels
/api/v1/media	File uploads
/api/v1/admin	Institution administration
/api/v1/superadmin	Platform administration

27. Database Architecture

27.1 Core Database Entities

Entity	Purpose
tenants	Institution records
users	User identities
groups	Academic groups
messages	Chat messages
media_files	Uploaded files
polls	Voting systems
audit_logs	Governance tracking
notifications	User alerts
sessions	Device sessions
ai_events	AI processing history

28. Performance Requirements

28.1 Performance Targets

Metric	Target
Message latency	< 300ms
API response	< 500ms
Concurrent WebSockets	10,000+ per node
OTP delivery	< 30 seconds
Upload processing	< 5 seconds
Platform uptime	99.5%

29. Future Expansion Strategy

29.1 Future Roadmap

Planned Expansions

- LMS integration
 - AI tutors
 - Assignment submission engine
 - Academic calendar synchronization
 - Attendance biometrics
 - Learning analytics
 - Institutional ERP integration
 - AI recommendation engine
 - Research collaboration tools
 - Marketplace ecosystem
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30. Commercial Vision

30.1 Long-Term Vision

UniChat Enterprise is designed not merely as an academic project, but as a scalable East African EdTech SaaS ecosystem capable of serving:

- universities,
- vocational colleges,
- training institutes,
- NGOs,
- and enterprise learning environments.

The platform architecture supports:

- regional expansion,
 - commercial SaaS deployment,
 - institutional white-labeling,
 - and long-term scalability.
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31. Conclusion

UniChat Enterprise represents a modern cloud-native academic communication ecosystem built upon:

- distributed systems principles,

- event-driven architecture,
- scalable real-time infrastructure,
- AI-enhanced workflows,
- and enterprise-grade SaaS design.

The platform combines:

- academic governance,
- communication infrastructure,
- institutional branding,
- and modern collaboration capabilities

into a unified digital ecosystem optimized for the educational environments of East Africa.

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